

Team Infinity

Problem Statement

In a digital landscape where advertising spending is forecasted to reach a staggering \$836 billion by 2026, a significant portion of this investment is jeopardized by digital ad fraud. Despite the growth in ad budgets, fraudulent activities like click fraud, domain spoofing, and pixel stuffing siphon off up to \$120 billion annually, leading to inflated performance metrics, wasted ad spending, and distorted campaign effectiveness. For Instagram, this threatens advertisers' financial integrity and undermines user trust and the platform's reputation as a reliable advertising medium.

Why Should We Solve This Problem?

Addressing digital ad fraud is crucial for several reasons:

1. **Financial Protection:** With advertisers losing a substantial part of their budgets to fraud, ensuring that ad spending is used effectively will protect financial investments and enhance ROI for brands using Instagram.
2. **Enhanced Trust:** By implementing a robust ad verification system, Instagram can restore and build trust among advertisers. Demonstrating commitment to combating fraud will position Instagram as a leader in ethical advertising.
3. **Market Competitiveness:** As digital ad fraud becomes more sophisticated, staying ahead with advanced verification technologies will differentiate Instagram from competitors, attracting more advertisers and boosting platform growth.
4. **Improved User Experience:** Reducing fraudulent ads will enhance the user experience on Instagram, ensuring that users see relevant and authentic advertisements, thereby increasing engagement and satisfaction.

Objective and Motivation

Objective

The primary objective of developing and implementing an AI-driven ad verification system on Instagram is to combat digital ad fraud by: effectively

- **Detecting Fraudulent Activities:** Identifying and mitigating fraudulent interactions such as click fraud, impression fraud, and domain spoofing.
- **Protecting Advertisers:** Ensuring that advertising budgets are spent on genuine user interactions and effective impressions.
- **Enhancing Ad Quality:** Improving the quality and relevance of ads shown to users, thus increasing user satisfaction and engagement.

- **Boosting Platform Integrity:** Reinforcing Instagram's position as a trusted and reliable platform for digital advertising.

Motivation

The motivation behind this initiative is to safeguard advertisers' financial interests, maintain the advertising ecosystem's integrity, and deliver a superior user experience. As digital ad fraud continues to evolve and impact the industry, proactive measures are essential to mitigate these threats and secure Instagram's role as a leading advertising platform.

Decoding the Existing System

How Ads Run on Instagram

Ads on Instagram are orchestrated through the Instagram Ads Manager, where advertisers create various ad formats such as photo, video, carousel, and story ads. The targeting mechanism on Instagram is sophisticated, allowing advertisers to reach specific demographics, interests, and behaviors. Advertisers also have the option to use Custom and Lookalike Audiences to refine their reach.

Once created, ads are strategically placed across different parts of the Instagram platform, including users' feeds, stories, the Explore tab, and IGTV content. The placement and visibility of these ads are governed by an auction system where advertisers bid for ad space. They manage their budgets on a daily basis or over the entire campaign period. Instagram employs machine learning algorithms to optimize ad delivery based on performance metrics such as impressions, clicks, and conversions, ensuring that ads reach the most relevant audience.

Retargeting on Instagram

Retargeting on Instagram involves displaying ads to users who have previously interacted with an advertiser's content or website. This process is facilitated by the Facebook Pixel, a tool that tracks user behavior and allows advertisers to create Custom Audiences for retargeting purposes. Dynamic Ads play a significant role in this strategy by showing products to users based on their previous interactions with the advertiser's site.

To ensure effective retargeting, advertisers can control ad frequency and segment their audience based on their interactions with the brand. Measurement and analytics tools are used to monitor the performance of retargeted ads, providing insights that help advertisers refine their campaigns and achieve better results.

THE ALGORITHM

Instagram’s ad algorithm is a sophisticated system that incorporates several critical factors to determine ad placement and effectiveness.

- The algorithm evaluates the relevance of an ad to a user. This relevance is assessed by analyzing the user’s historical interactions on the platform, such as likes, comments, and searches. Ads that align closely with a user's interests and behaviors are more likely to be displayed to them. The goal is to ensure that users encounter ads that are meaningful and engaging, thereby enhancing their overall experience.
- In addition to relevance, the algorithm considers the ad's performance metrics. These include click-through rates, engagement rates, and conversion rates. Ads that show strong performance in these areas are prioritized, while those with lower performance metrics may be shown less frequently or not at all. This performance-based approach helps in delivering ads that have a higher likelihood of resonating with users.
- Finally, the algorithm incorporates the advertiser’s bid into the equation. Advertisers can bid for ad placements, and higher bids generally lead to more prominent visibility. However, the bid alone does not guarantee top placement. The relevance and performance of the ad also play crucial roles in determining its visibility. This balanced approach ensures that users see ads that are both relevant and high-quality, while advertisers get opportunities to reach their desired audience effectively.

As Instagram continues to refine its platform, the algorithm is expected to evolve, providing even more tailored ad experiences for users and enhanced opportunities for advertisers.

Platform	Total Number of Users	Annual Ad Revenue
Instagram	2.5 billion	\$44 billion
Facebook	3 billion	\$114 billion
TikTok	1.6 billion	\$13 billion
Twitter	450 million	\$5.1 billion
LinkedIn	900 million	\$16 billion

This table provides a snapshot of the user base and advertising revenue for each platform, highlighting Instagram's significant ad revenue compared to its user base.

INDIAN CONSUMER BEHAVIOUR DATA

Metric	Data
Viewing Percentage	85% of Indian online users view ads
Buying Percentage	45% of Indian online users make purchases through ads
Geographical Data	
- Urban	65% of online ad viewers are from urban areas
- Rural	35% of online ad viewers are from rural areas
Average Conversion Time	
Domain: E-commerce	
- Low Ticket Size (< ₹1,000)	3-5 days
- Medium Ticket Size (₹1,000 - ₹10,000)	7-14 days
- High Ticket Size (> ₹10,000)	14-21 days
Domain: Travel & Hospitality	
- Low Ticket Size (< ₹5,000)	5-7 days
- Medium Ticket Size (₹5,000 - ₹20,000)	10-20 days
- High Ticket Size (> ₹20,000)	20-30 days
Domain: Consumer Electronics	
- Low Ticket Size (< ₹5,000)	7-10 days
- Medium Ticket Size (₹5,000 - ₹25,000)	15-25 days
- High Ticket Size (> ₹25,000)	25-35 days
Domain: Financial Services	
- Low Ticket Size (< ₹10,000)	5-10 days
- Medium Ticket Size (₹10,000 - ₹50,000)	10-20 days
- High Ticket Size (> ₹50,000)	20-30 days

Indian Consumer Behaviour

How Scammers Exploit Instagram for Fraudulent Activities

- **Targeted Advertising Abuse:** Scammers use Instagram's advanced targeting features to run ads that offer products or services at unrealistic prices. These ads often lead to fake websites designed to steal payment information or personal details.
- **Retargeting Techniques:** Once a user interacts with a fraudulent ad, scammers utilize retargeting to repeatedly show the same or similar ads, increasing the likelihood of scamming the user through persistent exposure.
- **Deceptive Content:** Fraudsters create high-quality, visually appealing posts and Stories that promote counterfeit or misleading products. This professional appearance can deceive users into believing the offers are genuine.
- **Influencer Manipulation:** Scammers may use fake accounts or compromised influencer profiles to endorse their fraudulent products. The perceived credibility of these endorsements can make the scams appear legitimate.
- **Weak Ad Review Process:** Instagram's ad review process may sometimes fail to catch fraudulent ads immediately. This delay allows scammers to exploit users and achieve their objectives before the ads are removed.
- **Visual Appeal for Legitimacy:** The platform's focus on high-quality images and engaging content helps scammers present their fraudulent offers in a convincing manner, making it harder for users to identify scams.

Core Problems Allowing Spam Ads on Instagram

- **Weak User Reporting Mechanism:** Users may find it difficult to report suspicious or fraudulent ads effectively, leading to delays in identifying and removing spam ads from the platform.
- **Insufficient Ad Verification Processes:** Instagram's verification processes for advertisers may not be stringent enough, allowing new or unverified accounts to run ads without adequate scrutiny.
- **Limited Transparency in Ad Metrics:** Lack of transparency in how ad metrics are reported can make it harder for both users and advertisers to detect anomalies or suspicious activities associated with spam ads.
- **Poor Collaboration with Third-Party Ad Verification Services:** Instagram may not collaborate effectively with third-party ad verification services, reducing the platform's ability to independently verify ad authenticity.

Market Research

Primary Research

User Interview Insights

† Deatiled User Interview

User 1	Details
About the User	Relocating to Dubai; Searching for an apartment.
Pain Points	Experienced online ad fraud while trying to rent an apartment; Suspicious invoice formatting saved from the scam; Concerned about the authenticity of online listings.
Key Takeaways	Needs enhanced verification of ad and listing authenticity; Prefers clear, trustworthy visual cues in ads to prevent fraud; High awareness of potential scams in the rental market.

User 2	Details
About the User	Looking for a jacket; Encountered a too-good-to-be-true ad.
Pain Points	Ad featured a very cheap jacket with a seemingly legitimate website; Retargeting ads increased suspicion of the deal being a scam; Verified the website’s legitimacy before making a purchase.
Key Takeaways	Values transparency and verification of online ads; High scrutiny of deals that seem too good to be true; Seeks reliable methods to confirm ad authenticity before purchasing.

Secondary Research Secondary Research

Global Cost of Digital Ad Fraud

Digital ad fraud represents a substantial financial burden. In 2019, estimates for the total cost of digital ad fraud ranged between \$6.5 billion and \$19 billion. By 2022, programmatic display advertising, a significant component of digital ads, accounted for a \$15 billion market, covering 90% of all digital ads. This underscores the pressing nature of ad fraud within the industry. Projections for 2023 indicate that the cost of digital ad fraud could reach \$100 billion, with a trend of increasing by approximately \$10 billion annually in recent years.

Ad Fraud by Region

The impact of digital ad fraud varies significantly across different regions:

- **APAC:** The region faces a severe ad fraud issue, with costs estimated at \$17 million per day. The total losses from ad fraud are expected to rise from \$33 billion to \$75 billion over the coming years.
- **China:** Ad fraud accounts for a significant portion of ad spend, with losses amounting to \$11 million for every \$63 million spent on digital advertising. This translates to a fraud rate of 17%.
- **United States:** In the U.S., \$62 million out of \$407 million spent per user is lost to ad fraud, representing 15% of the total spend.
- **India:** For India, the estimated ad fraud losses are approximately \$1.5 million for every \$10 million spent, reflecting a 15% fraud rate.

Average Ad Spend versus Fraud by Region

Across various regions, the average ad spend and corresponding fraud losses are as follows:

- **United States:** With \$407 million spent per user, \$62 million is lost to fraud.
- **China:** \$63 million is spent per user, with \$11 million lost to fraud.
- **India:** Approximately \$10 million is spent per user, with around \$1.5 million lost to fraud.

Global Ads Data

- Digital ad fraud accounts for **\$1** for every **\$3** spent on digital ads
- **7%** of advertisers surveyed are ready to pay a premium of **11%** or more for certified human traffic
- The rate of fraudulent traffic on Android is **31%** and **25%** of iOS
- Fraud instances are **25%** lower for in-app advertising than for the web
- **36%** of display ad clicks are fraudulent
- **17%** of CTV impressions are fraudulent
- **11%** of search ad clicks are fraudulent or invalid

- Just **12%** of ad impressions paid for by brands can be accounted for (i.e. matched)
- Ad fraud by region: China **\$18.7** billion lost, North America – **\$2.6** billion, EMEA – **\$804** million
- **24%** of total web bots are used for fraud and theft
- It is estimated that about **43%** of traffic is invalid traffic

Competitor Analysis Detailed Competitor Analysis

Tool	Key Features	Pricing	Rating	Top Clients
Anura.io	Real-Time Fraud Detection, Detailed Analytics, Customizable Filters, API Integration	Custom Pricing	4.7/5	AdRoll, LendingTree
Bolster.ai	AI-Powered Detection, Automated Takedowns, Comprehensive Reports, Real-Time Alerts	Custom Pricing	4.6/5	IBM, Salesforce
CHEQ	Full-Funnel Protection, Advanced Bot Mitigation, Multi-Layered Security, Easy Integration	Custom Pricing	4.8/5	Outbrain, Wix
Integral Ad Science	Contextual Targeting, Ad Fraud Detection, Brand Safety, Viewability Analysis	Custom Pricing	4.5/5	Verizon, Unilever
DoubleVerify	Ad Fraud Detection, Brand Safety, Viewability Verification, Comprehensive Reporting	Custom Pricing	4.7/5	AT&T, Toyota

White Ops (Human Security)	Bot Mitigation, Account Takeover Prevention, Marketing Integrity, API Integration	Custom Pricing	4.6/5	The Trade Desk, Adobe
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Insights from Market Research

Problems and Challenges Identified

Algorithmic Vulnerabilities:

Instagram's algorithm, while sophisticated, is not foolproof. Fraudsters manipulate user behavior signals and create fake engagement metrics to exploit these vulnerabilities, making it challenging to detect fraudulent ads accurately.

Verification Challenges:

The high volume and speed of ad placements on Instagram make real-time verification difficult. The platform struggles to differentiate between legitimate and fraudulent ads, leading to potential scams slipping through the cracks.

Evolving Fraud Tactics:

Fraudsters continually adapt their methods to stay ahead of detection technologies. This includes using advanced techniques such as deepfakes and hybrid fraud campaigns, which combine multiple fraud strategies to evade detection.

User Trust and Safety:

The presence of fraudulent ads undermines user trust in the platform. Users who encounter scams may become wary of engaging with ads, reducing overall engagement and the effectiveness of legitimate advertising campaigns.

Financial Impact:

Advertisers face substantial financial losses due to fraudulent clicks and impressions, which distort key performance metrics and reduce return on investment (ROI). This financial drain affects the overall efficiency of advertising budgets and can deter advertisers from investing in the platform.

User Personas

Attribute	Marketing Manager	Small Business Owner	College Student
Name	Priya Sharma	Rajesh Kumar	Ananya Iyer
Age	35	50	19
Occupation	Marketing Manager	Small Business Owner	Marketing Major Student
Location	Mumbai, India	Pune, India	Bengaluru, India
Background	Bachelor's in Marketing, MBA, 10+ years in digital marketing	Bachelor's in Commerce, 25+ years in business, now focusing on digital marketing	Sophomore in college, part-time intern at a marketing firm
Goals	Ensure correct ad display, maximize ROI, maintain brand safety	Optimize ad spend, reduce fraud, grow online presence for business	Learn about digital marketing, gain practical experience
Challenges	Tracking and verifying ads, ad fraud, limited time	Understanding digital tools, staying updated with trends, ad fraud	Balancing studies and internship, understanding complex tools
Needs	Reliable ad verification, detailed reports, easy integration	Simple ad verification, basic reports, fraud detection	User-friendly tool, learning resources, basic fraud detection
Tech Savviness	Comfortable with digital tools, basic analytics	Limited tech knowledge, prefers simple tools	Familiar with social media tools, learning analytics

Detailed User persona:  User Persona AI Ad verification

Solutions

a. High-Level Solutions

The overall high-level solution for combating fake ads on Instagram involves implementing advanced AI and machine learning models for pre-go live and post-go live screenings. Pre-go live, the system uses NLP, CNN, and behavioral analysis to score ads and classify them into low, moderate, and high risk, with manual reviews for moderate-risk ads. Post-go live, continuous monitoring assesses CTR, user engagement, and comments using NLP, updating risk scores dynamically and conducting regular rescreenings. This comprehensive approach includes user reporting, account health monitoring, and feedback loops to continuously improve model accuracy and ensure a positive experience for legitimate advertisers

Pre-Go Live Ad Screening

1. **Data Collection and Preparation:** Gather historical ad data, label it as genuine or fake, and include external datasets, user reports, and platform-generated data.
2. **Feature Extraction:** Extract key features from ads, including:
 - **Text Analysis:** Keyword frequency, sentiment, text length, spelling, and grammatical errors.
 - **Image Analysis:** Logo detection, image quality, visual errors, and manipulated images.
 - **Behavioral Analysis:** Account age, followers-to-age ratio, ad posting frequency, and user engagement metrics.
 - **Metadata Analysis:** Advertiser verification status, ad spend patterns, and URL security.
3. **Model Training and Inference:** Use advanced AI models such as:
 - **NLP Models:** BERT or GPT-3 for text analysis.
 - **CNN Models:** ResNet or EfficientNet for image and logo analysis.
 - **Anomaly Detection:** Isolation Forest or One-Class SVM for behavioral analysis.
 - **Ensemble Models:** Combine outputs from NLP, CNN, and anomaly detection models.
4. **Risk Scoring:** Develop a risk scoring engine to aggregate model outputs and classify ads into low, moderate, and high risk.
5. **Manual Review Process:**
 - **Moderate Risk Ads:** Manual reviewers check content authenticity, image verification, and advertiser history.
 - **Feedback Loop:** Integrate manual review findings into the model to improve accuracy.
6. **User Feedback Mechanism:** Notify advertisers of flagged ads with reasons and guidelines for improvement, and allow appeals for taken-down ads.

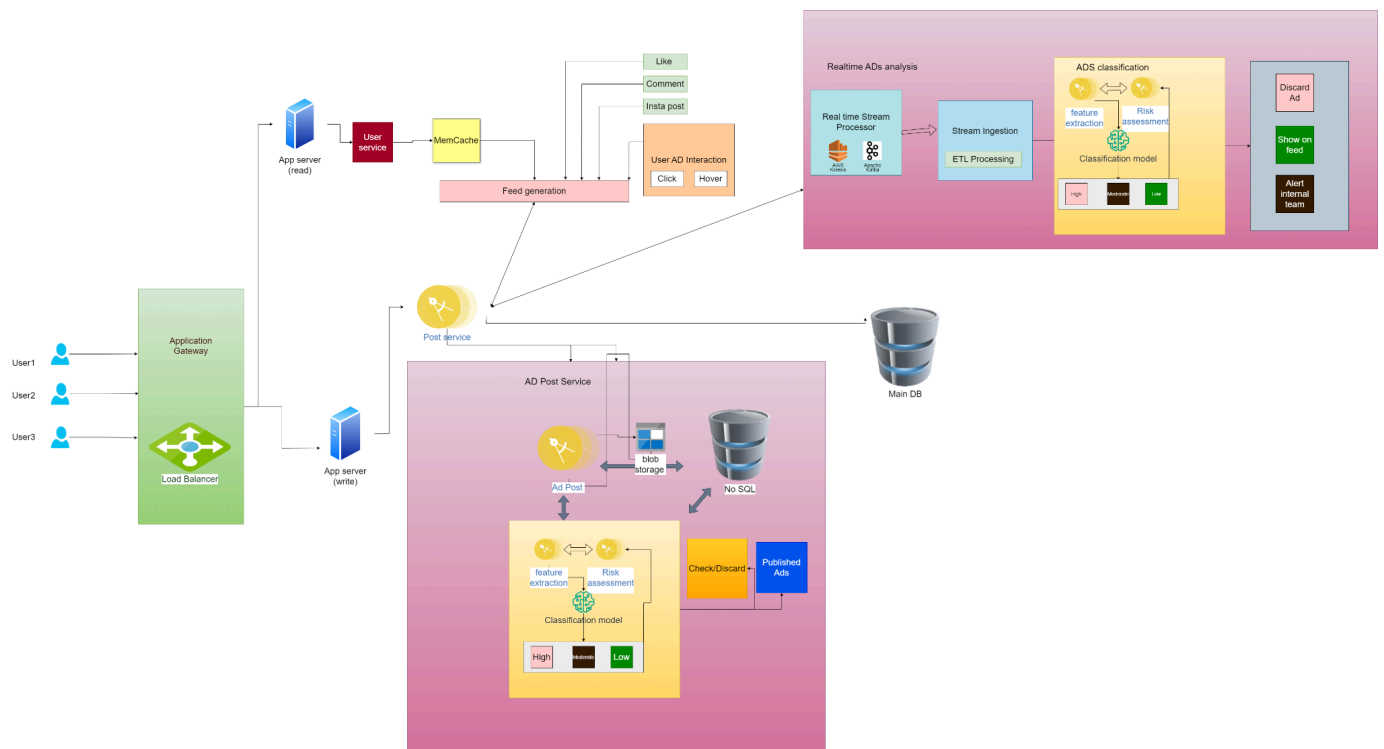
Post-Go Live Ad Screening

1. **Continuous Real-Time Monitoring:** Implement systems for continuous real-time screening of ads immediately after they go live.
2. **Advanced Behavioral Analysis:**
 - **Click-Through Rate (CTR) Analysis:** Monitor CTRs for unusual patterns.
 - **Engagement Metrics:** Analyze likes, shares, and comments.
 - **Ad Targeting Accuracy:** Measure audience matching accuracy.
3. **NLP-Based Comment Analysis:** Use NLP models to analyze comments for negative sentiments and suspicious keywords.
4. **User Reporting System:** Provide tools for users to report suspicious ads, with automated triage to prioritize reports.
5. **Dynamic Risk Scoring:** Continuously update risk scores based on real-time data, rescreening flagged ads and adjusting scores accordingly.
6. **Regular Rescreening Intervals:**
 - **Initial Screening:** Real-time for the first 24 hours post-go live.
 - **Scheduled Checks:** Daily for the first week, then weekly.
 - **Adaptive Screening:** Trigger-based rescreening for ads with suspicious activity.
7. **Account Health Monitoring:**

- **Behavior Analysis:** Monitor advertiser behavior and overall account health.
 - **Violation Policies:** Implement suspension and ban policies for repeated violations.
8. **Feedback and Learning:**
- **Continuous Learning:** Update AI models with new data from post-go live screenings and manual reviews.
 - **Advertiser Guidance:** Provide guidelines to advertisers to reduce the likelihood of their ads being flagged.
9. **Transparency and Communication:**
- **Advertiser Notifications:** Inform advertisers about the status and reasons for ad flags and takedowns.
 - **User Feedback Loop:** Provide feedback to users who report ads, informing them of the outcomes of their reports.

b. Detailed Solutions - [Team Infinity - AdSniffer - Detailed Solution](#)

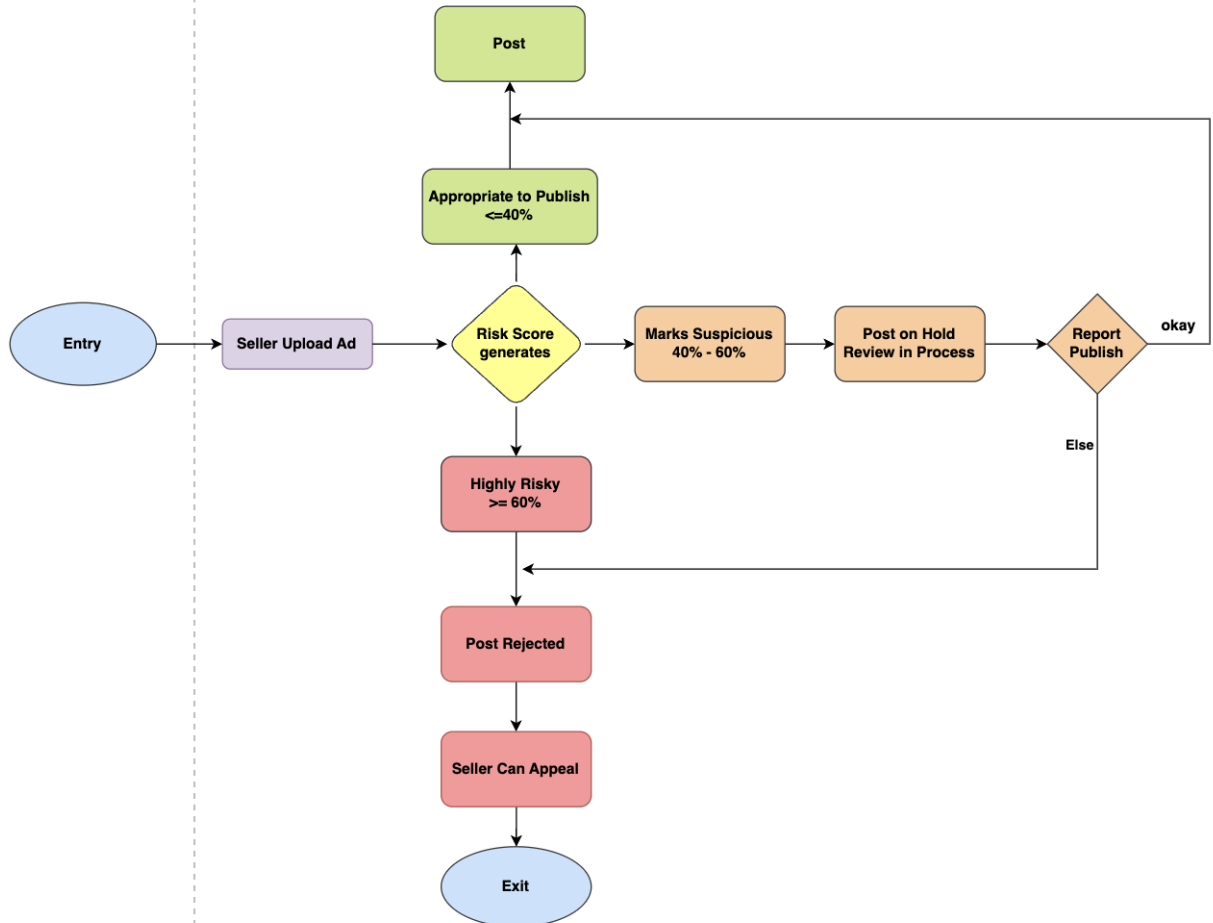
System Design



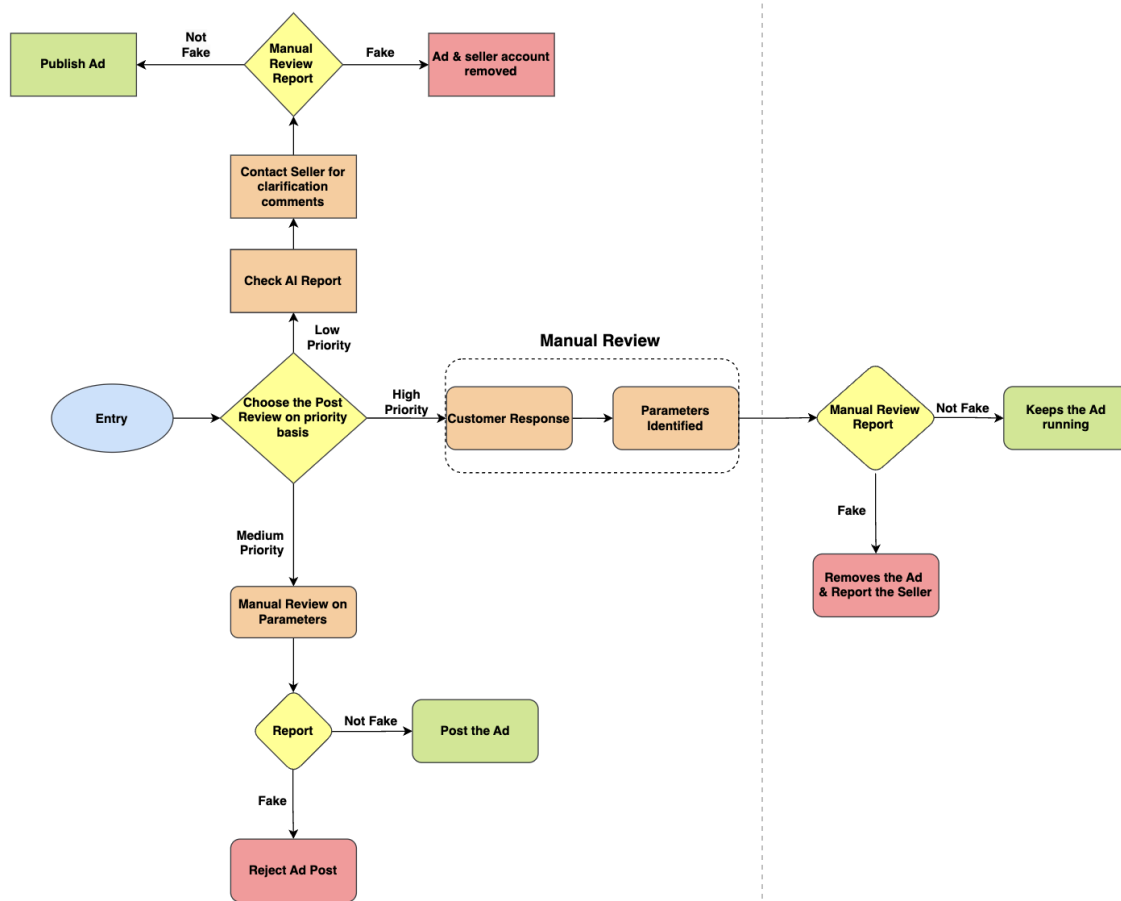
[Link to Detailed System Design](#)

User Flows

Seller Posting An Ad



Platform Response on a Fake Ad/ Platform User Flow



Detailed User Flow

Wireframes

User Reporting the Ad:

<https://www.figma.com/proto/gkFC97z7sQd2HU7VvkHp9i/Untitled?node-id=26-760&t=Nn1DQEcWqXYWd80-1>

<https://www.figma.com/design/gkFC97z7sQd2HU7VvkHp9i/Untitled?node-id=41-2223&t=Y2mYA6e1eOutJIY5-1>

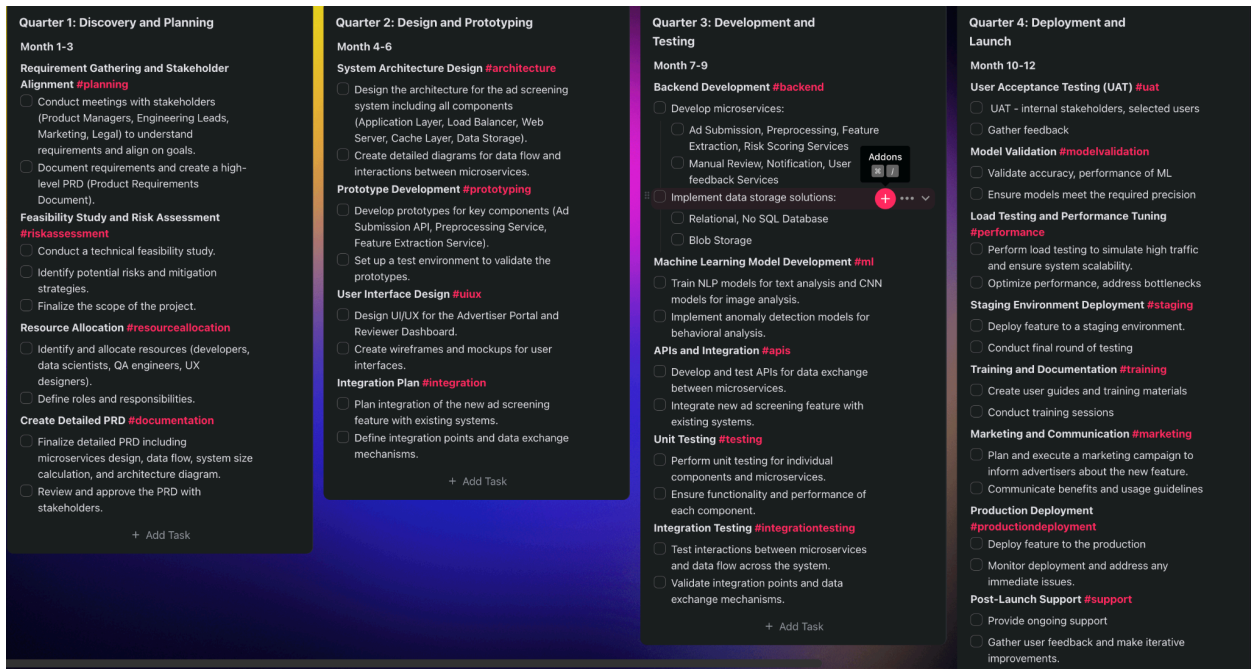
Metrics

North Star:

Accuracy Rate = {Number of Correctly Classified Ads (by AI and Manual Review) / {Total Number of Ads Processed}} * 100

 Team Infinity - Ad Sniffer - Metrics

Product RoadMap



[Link to Detailed Roadmap](#)

Go-To-Market (GTM) Strategy

Objective

Launch a comprehensive ad verification system, including pre go live and post go live ad screening, to enhance ad quality, maintain platform integrity, and provide a seamless experience for advertisers and users.

Target Audience

1. **Advertisers:** Businesses and individuals running ads.
2. **Review Teams:** Internal and outsourced ad verification teams.
3. **Platform Users:** General user base interacting with ads.

Key Messages

- **Advertisers:** Ensure ad quality, save time with reduced post-launch flags, transparent status updates, and increased trust in ad performance.
- **Users:** Safer browsing experience with fewer fake/misleading ads.
- **Internal Teams:** Streamlined ad review process, training, and support for accurate ad verification.

Marketing Channels

- **Email Campaigns:** Inform advertisers about new features, benefits, and usage guidelines.
- **Webinars/Workshops:** Demonstrate features, provide training, and address advertiser queries.
- **Content Marketing:** Publish blog posts, articles, and case studies showcasing early successes and best practices.
- **Social Media:** Share announcements, educational content, and live Q&A sessions.
- **In-App Notifications:** Guide advertisers through new processes and provide timely reminders.
- **Customer Support/Help Center:** Update help articles and ensure support teams are well-trained.

Implementation Plan

Quarter 1: Discovery and Planning

- Finalize PRD, conduct feasibility studies, and allocate resources.
- Develop detailed system design and architecture diagrams.

Quarter 2: Design and Prototyping

- Design system architecture and develop prototypes.
- Create UI/UX designs for Advertiser Portal and Reviewer Dashboard.
- Plan system integration with existing infrastructure.

Quarter 3: Development and Testing

- Develop backend microservices and frontend interfaces.
- Train and implement machine learning models.
- Perform unit, integration, and load testing.
- Conduct User Acceptance Testing (UAT) with selected advertisers.

Quarter 4: Deployment and Launch

- Deploy system in staging, then in production environments.
- Conduct training sessions and update support documentation.
- Launch marketing campaigns and in-app notifications.
- Monitor system performance and gather feedback for continuous improvement.

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